4 **(a)** Define *electric field strength*.

.....

.....

.....[1]

(b) An oil drop of mass 1.0×10^{-15} kg and carrying a charge of 1.6×10^{-19} C is placed in the middle of two horizontal parallel plates.

The separation between the plates is 40 mm and the the potential difference between the plates is 8.8 kV. The oil drop is seen to be attracted towards the upper plate because of its electric charge.

(i) Calculate the electric force experience by the oil drop.

force = ... N [2]

(ii) Initially the oil drop is 20 mm from either plate and is at rest. Ignoring air resistance, use energy considerations to calculate its speed when it reaches the upper plate.

speed = ... $m\ s^{-1}$ [2]

[Total: 5]